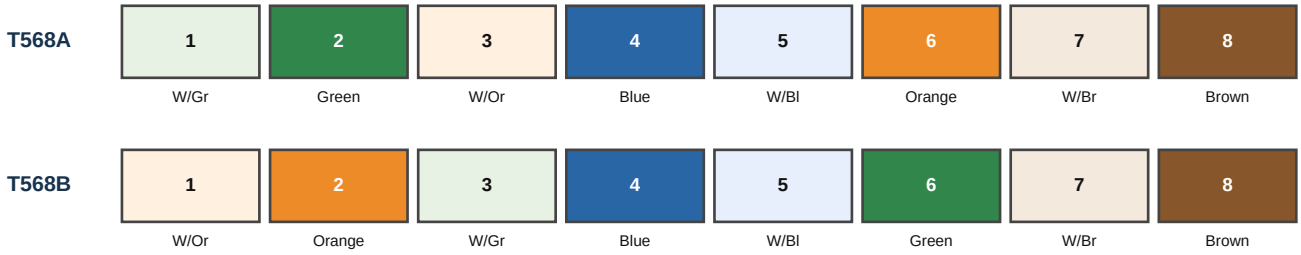
Ethernet Wiring and Standards Assignment

Name: _____ Class: _____

1. RJ-45 Pin Layout: T568A and T568B

An RJ-45 Ethernet plug has 8 pins (contacts). T568A and T568B are two wiring color standards. The pin numbers below are shown from left to right when looking at the gold contacts with the clip facing away/down.

RJ-45 Plug Pin Layout — T568A vs T568B



View the plug with gold contacts facing you and the clip on the back/bottom; pin 1 is at the left.

T568A order (pins 1–8): White/Green, Green, White/Orange, Blue, White/Blue, Orange, White/Brown, Brown.

T568B order (pins 1–8): White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.

2. Comparison of Ethernet Standards

Standard	Speed	Cable type	Maximum cable length	Connector
10BASE-T	10 Mb/s	Cat3 or better twisted-pair copper (commonly Cat5e/Cat6 today)	100 m per Ethernet link	8P8C modular plug, commonly called RJ-45
100BASE-TX	100 Mb/s (Fast Ethernet)	Cat5 or better, 2 twisted pairs	100 m per Ethernet link	8P8C modular plug, commonly called RJ-45
1000BASE-T	1000 Mb/s (1 Gb/s, Gigabit Ethernet)	Cat5 or better, 4 twisted pairs	100 m per Ethernet link	8P8C modular plug, commonly called RJ-45

Note: The 100 m limit usually means up to 90 m of fixed horizontal cable plus up to 10 m total patch cables in a standard channel.

3. Connecting Two Laptops Directly

Traditionally, a crossover Ethernet cable is used to connect two laptops directly because it swaps the transmit and receive wire pairs: one laptop's transmit pair connects to the other laptop's receive pair, and vice versa. A straight-through cable has the same wiring standard at both ends (for example, T568B at both ends).

However, most modern laptops and Ethernet adapters support Auto-MDI/MDI-X, which automatically adjusts transmit and receive pairs. Because of this, a straight-through cable often works for a direct laptop-to-laptop connection today. If the devices are older or Auto-MDI/MDI-X support is unknown, a crossover cable is the traditional safe choice.

4. Three Real-World Devices That Use Ethernet

Device	Likely standard	Reason
Home Wi-Fi router LAN port	1000BASE-T (Gigabit Ethernet), often; some ports may be 100BASE-TX	Connects computers, TVs, and other devices in the home network. Check the port label/specif
College desktop PC / lab computer	1000BASE-T, commonly	Modern PCs usually have Gigabit Ethernet for fast file transfers and internet access.
Network printer or POS terminal	100BASE-TX or 1000BASE-T	Many printers and payment/POS terminals use 100 Mb/s; newer models may support G

Conclusion: T568A and T568B define the wire color order in an Ethernet plug. 10BASE-T, 100BASE-TX, and 1000BASE-T provide 10 Mb/s, 100 Mb/s, and 1 Gb/s speeds respectively. Cable needs depend on the devices and their supported Ethernet features.